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MAZENE AMEUR

PhD Researcher in Computer Science
AI/ML for Zero-Touch Management of 5G/6G Networks

Research Profile

PhD researcher at EURECOM and Sorbonne University working on AI/ML-driven automation for 6G network management and orchestration. My work bridges cutting-edge research and real-world deployment through solutions aligned with leading industry standards, including 3GPP, ETSI and TM Forum. Author of 8 peer-reviewed publications in venues such as IEEE INFOCOM, IEEE ICC and IEEE Communications Magazine, and an active contributor to three flagship European 6G projects.

Education

PhD in Computer Science 2024 – November 2026
Sorbonne University, Paris, France · Research conducted at EURECOM, Sophia Antipolis
Thesis: Novel AI/ML Techniques for Zero-Touch Service Management in 6G Networks

MSc in Networking and Distributed Systems 2021 – 2023
University of Laghouat, Algeria · Ranked 1st in cohort
Thesis: Machine Learning for Resource Allocation in Partially Observable Networks (18.25/20)
GPA: 15.44/20

BSc in Information Systems 2018 – 2021
University of Laghouat, Algeria · Ranked 1st in cohort
GPA: 16.36/20

Baccalaureate, Experimental Sciences 2017 – 2018
1st November 1954 High School, Laghouat, Algeria · GPA: 17.02/20
Ranked among the top 10 students at state level in the national examination.

Research and Professional Experience

Research Engineer, European 6G Research Projects 2024 – present
EURECOM, Sophia Antipolis, France · 6G-BRICKS, SUNRISE-6G, 6G-DALI

- Developed an end-to-end **MLOps platform** for EURECOM's 6G facility using MLflow, MinIO and Kubernetes, enabling scalable model training, deployment, monitoring and lifecycle management.
- Built **LLM-powered AI agents** for intent-based networking, automating network configuration and optimisation from high-level user intents.
- Designed frameworks for **AI governance and compliance**, addressing transparency, fairness and regulatory adherence in AI-driven network management.
- Collaborated with industry partners (IBM Ireland, Lenovo Deutschland, Siemens) and academic partners (Sorbonne Université, University of Oulu) to align deliverables with project goals.
- Supervised PhD students and interns on **Intent-based Networking (IBN)**, **MLOps for 5G/6G core networks** and relevant **3GPP and TM Forum** standards, providing technical guidance, code review and support for publications and project deliverables.

Teaching Assistant Feb. – Jun. 2025
EURECOM · Introduction to Networks (engineering school), 5G Core Network (master's level)
Ran laboratory sessions, graded assignments and supported students with course material.

Peer Reviewer 2024 – present
IEEE Transactions on Mobile Computing, *IEEE Communications Magazine*, *IEEE IoT Magazine IEEE NFV-SDN 2025*, *IEEE CAMAD 2024*, *IWCMC 2025*.

Research Intern Feb. – Jun. 2023
XLIM Laboratory, Poitiers, France · 4.5 months
Studied state-of-the-art DRL solutions for path selection in networking and proposed a novel approach. Conducted as part of the master's graduation project.

Selected Publications

Journal Articles

[1] **M. Ameur**, B. Brik and A. Ksentini. "Dual Self-Attention is What You Need for Model Drift Detection in 6G Networks." *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 3, pp. 690–709, 2025. doi:10.1109/TMLCN.2025.3576727.

[2] **M. Ameur**, B. Brik and A. Ksentini. "EUROCOMPLY: Enabling Zero-Touch AI Compliance Auditing via LLM-based Agentic AI." *IEEE Communications Magazine*. doi:10.1109/MCOM.001.2500366.

Conference Papers

[3] **M. Ameur**, B. Brik and A. Ksentini. "Uncertainty-aware Zero-Touch Drift Management for Trustworthy DRL in 5G/6G Networks." *IEEE INFOCOM 2026*, **Core A***, Tokyo, Japan, 2026.

[4] **M. Ameur**, B. Brik and A. Ksentini. "Agentic-NWDAF: Enabling Intent-driven Agentic Intelligence for Autonomous 6G Network Analytics." *IEEE ICC 2026*, Glasgow, UK, 2026.

[5] **M. Ameur**, B. Brik and A. Ksentini. "Adapt but Do Not Forget: Towards Enhancing Drift Handling in 6G Networks." *IEEE ICMLCN 2025*, pp. 1–6, 2025.

[6] **M. Ameur**, A. Bradai and N. Lagraa. "Exploring Teacher-Student Learning with Multi-Agent DRL for QoS Routing in SDN." *IEEE ICC 2025*, 2025.

[7] **M. Ameur**, B. Brik and A. Ksentini. "Leveraging LLMs to Explain DRL Decisions for Transparent 6G Network Slicing." *IEEE NetSoft 2024*, pp. 204–212, 2024.

Demonstrations

[8] **M. Ameur** and A. Ksentini. "When MLOps Meet NWDAF Release 18." Demonstration, *ETSI SNS4SNS 2026*.

[9] **M. Ameur** and A. Ksentini. "Automated Regulatory Compliance Validation for AI/ML Systems in Telecommunications via Agentic AI" Demonstration, *ETSI AI & Data Conferece 2026*.

[10] **M. Ameur** and A. Ksentini. "EUROCOMPLY: Enabling Automated Regulatory Compliance Auditing AI Systems" Demonstration, *EuCNC 2026*, Link: https://www.youtube.com/watch?v=nczcZeryCxM

Selected Projects

OAI-NWDAF · **Open Source Maintainer and Contributor** 2025 – present

OpenAirInterface NWDAF upgrade to 3GPP Release 17/18 compliance · gitlab.eurecom.fr/netsoft/nwdaf-rel17
Architected and integrated cloud-native, service-based MLOps capabilities within the NWDAF core, enabling the transition from Release 15 to Release 17/18. Led the refactoring that enforces strict logical separation between the Analytics Logical Function (AnLF) and the Model Training Logical Function (MTLF), in line with 3GPP specifications.
Extended the platform with horizontal Federated Learning across multiple NWDAF instances, enabling collaborative model training between distributed MTLFs without sharing raw data. Federated-Learning branch

6G-DALI · **6G Data and ML Operations Automation via an End-to-End AI Framework** 2024 – Present

Tailored MLOps capabilities for the NWDAF core network, led its upgrade from 3GPP Release 15 to Release 18, and strengthened AI governance and compliance mechanisms.

SUNRISE-6G · **Federation of Research Infrastructures for 6G Experimentation** 2024 – Present

Upgraded EURECOM's 6G testbed with a Kubernetes-based MLOps platform supporting scalable model training, deployment and experiment lifecycle management.

6G-BRICKS · **Reusable Testbed Infrastructures for Cloud-to-Device Technologies** 2024 – 2025

Applied LLMs to explain DRL decisions in network slicing use cases.

DRL-based Framework for Path Selection in SD-WAN 2023

Master's graduation project with XLIM Laboratory, Poitiers, France. Supervised by Prof. Abbas Bradai (University of Poitiers) and Prof. Nasreddine Lagraa (University of Laghouat).

Minimising Energy Consumption Using AI 2022

Applied Q-learning to reduce the energy consumption of an industrial installation.

Technical Skills

Standards & Regulation	3GPP, ETSI, TM Forum, EU AI Act, GDPR
ML & AI	Deep Learning, Reinforcement Learning, Federated Learning, LLMs, Agentic AI, XAI
MLOps & Cloud	MLflow, MinIO, Kubernetes, Docker, Ollama, CrewAI, ChromaDB, FastAPI, Flask, n8n
Programming	Python, Java, OMNeT++, NS-2, Mininet, Android Studio, Scilab
Databases	PostgreSQL, MySQL, MongoDB, SQL Server
Tools	Linux, LaTeX, Microsoft Office, OpenOffice, Adobe Illustrator, Adobe Photoshop
Languages	English, French, Arabic (full professional proficiency)

Certifications

- **MLOps Bootcamp: Mastering AI Operations for Success (AIOps)** · Udemy, 2024
- **Explainable AI (XAI) with Python** · Udemy, 2023
- **Deep Learning A-Z: Hands-On Artificial Neural Networks** · Udemy, 2022
- **EnglishScore Skills Test, level C1** · British Council, 2021

Research Interests

- Agentic AI, large language models, deep learning, federated learning, reinforcement learning, explainable AI.
- Zero-touch service management, network management, software-defined networking, 5G/6G networks.
- Telco, cloud and edge computing, UAV networks, vehicular networks.

References

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